



Care of Peripheral Midline Catheter Procedure

1. Scope

Site	Operational Area	Applicable to
Armada Kalamunda Group	All Clinical Services	Medical, Nursing, and Midwifery

2. Purpose

To provide guidance for safe care and management of Peripheral Midline catheters.

All Midlines are to be reviewed each shift and removed when no longer clinically indicated. Staff are asked to avoid taking blood pressure on the arm with the Midline due to risk of occlusion.

- Care and removal of Midlines will only be undertaken by persons with the appropriate training in the procedure, or under the direct supervision of an appropriately trained person.
- Insertion of Midlines will only be undertaken by the Clinical Nurse Specialist Vascular Access or a nurse suitably trained by the Clinical Nurse Specialist Vascular Access.

3. General Management

No chest x-ray is required for placement as Midline tip ends in the Axillary Vein.

- Midline dressings to be done weekly or PRN if wet, soiled or blood soaked. All
- components of the Midline except for the line are changed with every dressing change.
- Midlines are to be covered with a transparent IV dressing to protect the site from extrinsic contamination and allow 8 hourly observation and documented on the AKMR 121.1 PICC and Midline Nursing Care Plan.
- Midlines must not extend a dwell time of 28 days.
- Hand Hygiene must be done before careful handling of line.
- Scrub the hub for 15-30 seconds with a 70% alcohol swab and allow to dry for at least 30 seconds.
- Only use in-situ clamps when changing positive pressure valve (PPV). If a PPV is in place leave clamp open.
- Flush the line pre and post ALL infusions/injections using 10mls Sodium Chloride 0.9% (use prefilled syringe where available). Use a push-pause method to create turbulence in the line maintain patency. Do not bottom out the flush syringe to end on positive pressure within the catheter.
- Flush the Midline with a push-pause method immediately after any blood sampling, with 20ml sodium chloride, using the push pause method to increase the turbulence in the line and remove any build-up of particulate or blood.
- All Midlines should be fitted with a Positive Pressure Valve (such as BD Maxplus clear needleless connector) to prevent line blockage.

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- Blood products can be transfused via Midline but must be flushed with 20mls of Sodium Chloride 0.9% post infusion.
- If Midline is blocked or leaking do not remove - contact the Clinical Nurse Specialist Vascular Access immediately.
- Cover Midline dressing with waterproof cover when showering and change dressing if wet or soiled.
- Do not use syringes smaller than 10ml as this may cause excessively high internal pressures or line rupture.

4. Dressing changes

- Dressing changes are to be done routinely every 7 days.
- Dressing changes are to be done immediately if dressing integrity is compromised.
- The catheter insertion site should be visually inspected and palpated through the intact dressing to determine tenderness during 8 hourly review. The process of dressing changes, when indicated, provides the opportunity closely review and cleanse the insertion site.
- Suture less securement device if in-situ requires changing every 7 days.
- Cover the dressing with waterproof cover when showering

4.1 Equipment

- 10ml prefilled Sodium Chloride 0.9% syringe
- Sterile transparent semi-permeable IV Dressing with Chlorhexidine gel such as 3M Tegaderm CHG OR Chlorhexidine gluconate impregnated foam disk and plain semi-permeable IV Dressing.
- Antimicrobial swab sticks with 70% alcohol 2% chlorhexidine (e.g. Reynard Swabsticks or 3M SoluPrep Antiseptic Swab) OR antimicrobial Chlorhexidine Ethanol surgical prep solution with 70% alcohol 2% chlorhexidine.
- Skin barrier film swab stick (e.g. 3M Cavilon No sting barrier film).
- PPE: sterile & non-sterile gloves.
- Positive Pressure Valve (e.g. Maxplus clear).
- Sterile dressing pack
- Suture less securement device (e.g. 3M Securement Device- comes with 3M Tegaderm CHG dressing).
- Tubular bandage to cover limb.

4.2 Procedure

- Perform hand hygiene, clean work surface and assemble equipment.
- Position patient to expose Midline insertion site.
- Set up sterile dressing pack and open sterile equipment onto aseptic field (PPV, dressing, skin barrier film, surgical prep solution [if used], antimicrobial disc [if used]).
- Perform hand hygiene, don nonsterile gloves.
- Remove transparent dressing, peeling slowly from distal to proximal end to avoid dislodging Midline. If fragile skin is apparent, consider using adhesive removal swab.
- Remove suture less securement device. Ensure Midline is stable and do not leave the patient unattended.
- Clean site with antimicrobial swab sticks using aseptic non-touch technique (ANTT) until all visible debris is removed. Allow to air dry. Alternatively: Perform hand hygiene, don sterile

gloves. Clean Midline insertion site with antiseptic surgical prep solution until all visible debris is removed. Allow to air dry.

- Clean entire area that will be covered by dressing.
- Apply skin barrier film (e.g. 3M Cavilon No sting barrier film). Allow to dry.
- Replace suture less securement device, as per product information.



- Apply transparent IV Dressing with Chlorhexidine gel. Ensure gel pad covers insertion site. Write the date that the dressing changed. Alternatively: Place chlorhexidine gluconate impregnated foam disk (blue side up and white foam side down). Ensure split in disc lines up with catheter and makes full unbroken contact with skin. Apply transparent IV dressing over site. Write date of dressing performed.
- Prime PPV with Sodium Chloride 0.9%, remove used valve and attach primed valve to line and flush with Sodium Chloride 0.9% in a 10ml syringe using the PUSH/PAUSE technique.
- Apply tubular bandage over dressing if appropriate.
- Record date of dressing change, length of exposed catheter and next dressing due on AKMR 121.1 PICC and Midline Nursing Care Plan.
- Dispose of waste. Remove gloves and perform hand hygiene.

5. Midline flushing

Midlines are to be flushed pre and post every use. Flush with at least 9ml of Sodium Chloride 0.9% in a 10ml syringe. Flush with at least 20mL of saline post blood product administration or blood draw.

5.1 Equipment

- 10mL prefilled Sodium Chloride 0.9% syringe.
- 70% alcohol wipe.

5.2 Procedure

- Scrub the PPV hub for 15-30 seconds with 70% alcohol wipe.
- Allow to dry for at least 30 seconds.
- Flush line with a push-pause method. Do not forcibly administer flush if resistance encountered.
- Administer medications.
- Flush with at least 9mL of saline push-pause method to create turbulence within the catheter.
- Do not bottom out syringe when flushing to end flush on positive pressure.

6. Changing Positive Pressure Valve (PPV)

Replace PPV every 7 days with dressing changes, if contaminated, if unable to flush clear blood cells post blood draw, or leaking.

6.1 Equipment

- 10mL prefilled Sodium Chloride 0.9% syringe.
- Non-sterile gloves.
- PPV.

6.2 Procedure

- Perform hand hygiene.
- Attach prefilled syringe to the new sterile PPV without contaminating the PPV and prime it. Leave 10mL syringe attached.
- Clamp Midline.
- Remove and discard existing PPV.
- Attach new PPV.
- Remove clamp and flush Midline with a push-pause technique. Repeat for each lumen if applicable.
- Discard equipment and perform hand hygiene.
- Document details of procedure.

7. Midline blood sampling

7.1 Indications for blood sampling

- For patients where no other alternative exists.
- Difficult venous access.
- Following placement of a new device if clinically indicated.
- To investigate suspected line related blood stream infection.

7.2 Equipment

- Vacutainer blood tubes.
- Pink vacutainer safe blood transfer device.
- 2x10ml luer-lock syringes.
- 20ml luer-lock syringe.
- 70% alcohol wipe.
- Kidney dish.
- Non-sterile gloves.
- Sodium Chloride 0.9% for injection 3x 10ml minimum (for flushing).
- Specimen request form.
- For blood cultures:
 - 20ml luer-lock syringe
 - 2x vacuumed aerobic and anaerobic blood culture specimen bottles (one set collected from existing device and one set collected from separate peripheral venepuncture)

7.3 Procedure, general blood sampling

- Confirm correct patient details with pathology request form.
- Perform hand hygiene, clean work surface and assemble equipment.
- Cease infusion if in progress and if clinically appropriate.
- Perform hand hygiene. Don gloves.
- Utilise the largest lumen available.
- Undertake blood sampling from lumen with no continuous infusion in progress.
- Scrub the PPV hub for 15-30 seconds with a 70% alcohol wipe and allow to dry for at least 30 seconds.
- Flush line prior to blood sampling and always follow blood sampling policy.
- Attach empty 10ml luer-lock syringe onto PPV and withdraw 5 ml blood.
- Discard syringe containing blood.
- Withdraw the required amount of blood using either a 10ml or a 20ml luer-lock syringe.
- Place syringe with blood sample into the kidney dish.
- Flush all lumens with at least 20 mL sodium chloride 0.9% using a pulsatile method (push-pause) until lumen and PPV are clear.
- Transfer blood into appropriate vacutainer tubes using pink vacutainer safe blood transfer device.
- Label samples with patient identification sticker, date/time sample taken.

7.4 Procedure, blood cultures

- Follow steps above.
- Scrub the PPV hub for 15-30 seconds with a 70% alcohol wipe and allow to dry for at least 30 seconds.
- Do not flush line or discard first draw.
- Attach 20ml luer-lock syringe and withdraw 20ml of blood.
- If more than one lumen, ensure to withdraw blood from all lumens.
- Place syringe with blood sample into the kidney dish.
- Flush all lumens with at least 20 mL sodium chloride 0.9% using a pulsatile method (push-pause) until all lumens and PPV are clear.
- Clean the top of the blood culture bottles with alcohol swabs.
- Attach pink vacutainer safe blood transfer device to syringe, place 10 mL of blood into each bottle. Label bottles as taken from MIDLINE.
- Ensure a second set of blood cultures are taken from a peripheral source. Label bottles as taken PERIPHERAL.

8. Midline removal

8.1 Considerations

When localised and/or systemic infection suspected, liaise with medical officer and obtain insertion site swabs for microscopy, culture, and sensitivity (MC&S). The device tip should also be cultured along with blood samples from the device along with peripheral blood samples.

- Send Midline tip for MC&S after removal, if infection suspected.
- Routine replacement of Midline is not recommended, unless inserted under conditions where strict asepsis was not maintained.

- Midlines require daily assessment and liaison between nursing and medical staff to determine continuing clinical need, removal when no longer required, or removal in situations of unresolved complications.

8.2 Equipment

- Nonsterile gloves.
- Adhesive removal solvent.
- Sterile gauze.
- Small occlusive dressing.
- If infection suspected:
 - Blood sampling equipment
 - Sterile dressing pack
 - Sterile scissors
 - Sterile yellow top specimen container

8.3 Procedure

- Position patient in a supine position, as tolerated.
- Position and support arm at 45-90 degrees.
- Perform hand hygiene. Prepare equipment, open sterile gauze.
- Don nonsterile gloves
- Complete blood sampling (if infection suspected).
- Carefully remove midline dressing. Use adhesive removal solvent if fragile skin present. Remove securement device.
- Obtain insertion site swabs (if infection suspected).
- Withdraw Midline with slow intermittent traction, without applying tension.
- If resistance is felt, stop removal, cover site with sterile dressing. Contact CNS Vascular Access.
- Place gauze over device removal site.
- Place removed line on sterile field, cut the tip off using ANTT and place in yellow top specimen container (if infection suspected).
- If infection is not suspected, discard line.
- On removal apply gentle pressure with gauze on the site until haemostasis is achieved.
- Apply sterile occlusive dressing and leave intact for a minimum of 24 hours.
- Continue PIVAS scoring of the site for 48 hours post removal.
- Document removal and label samples (if applicable).

9. Legislative context

The following documents are required to give effect to this procedure:

- [Infection Prevention and Management Policy EMHS: 34](#)
- [WA Health Consent to Treatment Policy](#)
- [Patient Identification and Procedure Matching](#)

10. Supporting information

The following documents support the implementation of this procedure:

- [Hand Hygiene Policy](#)

- [Aseptic Technique Policy](#)
- [Blood and Blood Product Management Guideline](#)
- [Specimen Collection and Pathology Results Procedure](#)

11. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 3 – Action 3.8
- Standard 3 – Action 3.9
- Standard 3 – Action 3.10

12. Glossary

The following definitions are relevant to this policy.

Term	Definition
Midline	A MIDLINE is a long Peripheral Catheter inserted via a vein in the upper arm with the tip residing in axilla vein. MIDLINES can be left in-situ up to 28 days.

13. Policy owner (relating to these procedures)

Director of Nursing and Midwifery

Enquiries relating to this procedure may be directed to:

Title: Shane Hastings
 Department: Nursing Coordinator
 Email: Shane.Hastings@health.wa.gov.au

14. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V.3.0	12/12/2024	12/12/2027	Scheduled review, minor amendments.

15. Authorisation

This procedure has been authorised and issued by the Director of Nursing and Midwifery.

Authorised by	Helen Fullarton – Director of Nursing and Midwifery
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